Revenue continued

Total Group revenue grew by $26.7 million in 2019, an increase of 5.6 per cent over the prior year.

The Networks & Security operating segment achieved year-on-year revenue growth of 12.2 per cent. Within that segment, all of our lines of business grew revenue compared to the prior year but the primary growth drivers were demand for high-speed Ethernet testing within our core business, particularly from China customers, and satellite simulators provided by our Positioning business, particularly US government related. The cybersecurity line of business also experienced revenue growth, although this remains a relatively small part of the overall Networks & Security portfolio.

Revenue at Lifecycle Service Assurance was essentially level year-on-year with revenue growth for next-generation automated validation and assurance solutions offset by a reduction in legacy solution revenues. However, we experienced robust growth in order intake year-on-year, growing the orderbook as a result of strong demand for our automated validation and assurance solutions for 5G, SD-WAN and virtual networks. New product offerings for automated NetDevOps testing and TaaS managed services also saw significant initial orders late in the year, with revenue to be recognised in 2020 and beyond.

Connected Devices experienced a reduction in revenue of 8.2 per cent, primarily a consequence of the continuing decline in the 4G legacy smartphone device lab test business outpacing the ramp-up in 5G-related business. Notwithstanding the fall in recognised revenue, order intake showed some growth over the prior year, as several large multi-year support contracts and orders closed in the final quarter that will not convert to revenue until 2020.

Geographically, we have continued to see growth in revenue from the Asia Pacific region (up 18 per cent year-on-year), particularly China customers purchasing high-speed Ethernet test solutions, with the Americas broadly level and EMEA marginally down. The growth in China revenue has been achieved despite the ongoing geopolitical landscape challenges. However, we saw order intake growth in the Americas and EMEA and closed the year with a solid orderbook for all regions.

"2019 trading performance demonstrably proves our strategy is working with continued revenue growth and another year of material profit growth."

Gross margin

$ million	2019	%	2018	%
Networks & Security	**232.3**	**72.6**	205.3	72.0
Lifecycle Service Assurance	**88.6**	**79.7**	87.9	77.9
Connected Devices	**47.7**	**65.8**	51.3	64.9
	368.6	**73.2**	344.5	72.2

The Group achieved further gross margin expansion in 2019 with an increase of 1.0 percentage points, to 73.2 per cent from 72.2 per cent in 2018. This followed an increase of 0.7 percentage points last year. Once again, all the operating segments achieved an improvement in gross margin, benefiting from new product launches and a higher proportion of software sales.

Operating costs

$ million	2019	2018
Product development	**96.5**	96.9
Selling and marketing	**129.2**	123.9
Administration[1]	**50.0**	46.6
Adjusted operating costs[1]	**275.7**	267.4
Networks & Security	**158.4**	148.9
Lifecycle Service Assurance	**70.5**	70.5
Connected Devices	**38.2**	40.8
Corporate	**8.6**	7.2
Adjusted operating costs[1]	**275.7**	267.4

Note
1. Before exceptional items, acquisition related costs, acquired intangible asset amortisation and share-based payment amounting to $4.3 million in total (2018 $19.6 million).

Total Group adjusted operating costs were up $8.3 million or 3.1 per cent in 2019 compared to last year, broadly in line with inflation. The emphasis remained on effective resource allocation and careful cost management. The overall investment in product development was maintained, with continuing focus on high-growth, high-margin areas. Investment in the sales and marketing organisation was targeted on expanding our key account management programme to drive incremental business with our most valuable customers and developing routes to market for our new technologies to a broadening customer base. Administration costs in 2019 reflected an inflationary increase and higher corporate costs, primarily due to CEO transition.

Segmentally, investment continued in Networks & Security, where we see the most near-term opportunities for growth, particularly in relation to 400G high-speed Ethernet and our Positioning business. A new General Manager joined Lifecycle Service Assurance in October and a review is in progress to evolve the business and optimise the organisational structure to expand the customer base and deliver on our growth agenda. Proactive cost management has once again been demonstrated within Connected Devices, where we have seen a decrease in legacy product revenue year-on-year. As stated above, corporate costs in 2019 included costs associated with CEO transition.